BOUBAKAR DIALLO

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EDUCATION

Cornell University — **B.S. Electrical & Computer Engineering** — **May 2028**

Honors Community Leadership Award - Cornell University (Aug 2025)

Coursework Circuits, Digital Logic & Computer Organization, Computer Architecture, Embedded Systems, Signals & Systems, Computer Systems Programming

TECHNICAL SKILLS

Circuit & PCB Design Circuit Prototyping, Motor Drivers, Power Regulation, Power Distribution, Board Bring-Up

Digital Hardware Verilog/SystemVerilog, RTL Design, FPGA Prototyping, FSMs, Synthesis, Timing Analysis

Embedded Hardware ARM Cortex-M, GPIO, I^2C, SPI, UART, Interrupts, Sensor Interfaces, RTOS

Test & Validation Oscilloscopes, Logic Analyzers, Multimeters, Sensor Calibration, Hardware Debugging

Programming & Tools C/C++, Python, ARM Assembly, MATLAB, Vivado, Quartus, Linux, Git/GitHub

WORK EXPERIENCE

Embedded Systems Intern *ADAS SAFE, Jacksonville, FL* **Jun – Aug 2025**

- Validated automotive radar, IMU, and LiDAR hardware, isolating calibration and synchronization faults to reduce radar alignment errors by 15%.
- Diagnosed microsecond-level timing drift across perception sensors, characterizing hardware synchronization dependencies that informed sensor-fusion interface design.
- Supported production perception-hardware bring-up and validation, investigating sensor-interface failures and calibration inconsistencies.

Robotics & Hardware Systems Lead *Engineers Without Borders, Ithaca, NY* **Oct 2024 – Jan 2026**

- Designed and fabricated custom PCBs, motor-driver circuits, and power-regulation hardware for an agricultural rover adapted from NASA's open-source Mars rover architecture.
- Led board bring-up and hardware/software integration, debugging power distribution, actuator control, and sensor-interface faults using laboratory instrumentation.
- Integrated embedded computing and a ROS-controlled robotic arm with the rover's electrical and control subsystems.

Software Engineer Intern *Simulacrum, NYC, NY* **Jan – Jun 2026**

- Engineered a distributed real-time sensor-data platform using FastAPI and Google Cloud Pub/Sub, processing high-throughput time-series streams with sub-100 ms latency for forecasting and model-training workflows.
- Built observability pipelines with structured logging and automated alerts for latency spikes, data drift, and processing failures, reducing debugging cycles by approximately 30%.

Incoming Technology Intern *PNC Bank* **Summer 2027**

PROJECTS

FPGA Digital Logic & Timing Optimization **Dec 2025**

- Designed and synthesized gate-level and RTL 16-bit counters, reducing critical-path delay by 68.3% (24.553 to 7.774 ns) and FPGA utilization by 92.9% (1,545 to 109 ALUTs).
- Implemented a Moore FSM-based FPGA music player with programmable counters and note-generation logic, validating signal timing through oscilloscope measurements.
- Analyzed FPGA synthesis and timing reports to quantify area–performance tradeoffs, comparing RTL and gate-level implementations across critical-path delay and logic utilization.

RISC-V Processor & Hardware Accelerator **Dec 2025**

- Implemented a single-cycle TinyRV1 RISC-V processor with 32-bit Datapath, register file, ALU, instruction decode, and control logic.
- Designed a custom accumulation accelerator with dedicated address/sum datapaths and FSM control, achieving 4.09× cycle-count speedup over the processor baseline (131 to 32 cycles).

Hexacopter UAV Electrical Integration **Jul 2026**

- Integrated six ESCs, power-distribution hardware, GPS/compass, and Pixhawk flight controller with telemetry.
- Calibrated electrical and flight-control interfaces, configured low-battery and RC-loss failsafes, and validated system integration through bench and tethered tests.